UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES (UITS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DHAKA, BANGLADESH

CSE 416: Software Project Design and Development (Lab)

SOFTWARE TEST REPORT

The Round Table Exchange (RTE): A Multi-Party Skill Barter Network with Graph-Based Cycle Matching

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1. Introduction & Overview

1.1 Purpose & Product Architecture: This report describes the formal verification of RoundTable Exchange (RTE) v1.0.1-test according to ISO/IEC/IEEE 29119-3 and ISO/IEC 25010. RTE is a circular skill barter platform that eliminates the 'Double Coincidence of Wants' by finding closed trade cycles of length k in {2, 3, 4} on a directed multigraph using Bounded DFS. Discovered cycles are scored by a composite scoring function:
$S(C) = \alpha\ S_geo(C) + (1 - \alpha)\ S_sched(C)$ combining Haversine proximity and weekly schedules. Unanimous single-veto consensus.

1.2 Scope of Testing

Subsystem / Functional Area	SRS Requirements	Testing Levels
User onboarding & Dhaka coordinate profiles	FR-01 to FR-05, FR-24	Unit, Integration
Skill taxonomy & directed multigraph construction	FR-06 to FR-14	Unit, Integration
Bounded DFS cycle discovery (k in [2, 4])	FR-15 to FR-21	Unit, Algorithmic
Haversine proximity & schedule overlap scoring	FR-22 to FR-28	Unit, Integration
Atomic 5-stage proposal state machine	FR-29 to FR-33	Unit, Integration
Synthetic scaling, REST API & client parity	FR-34 to FR-46, NFR-01..16	Perf, Parity, Security

2. Test Environment & Strategy

Component	Specification & Tooling Configuration
Host OS & CI Container	Windows 11 Enterprise (local test harness); Linux x86_64 (CI container)
Language Runtimes & Core	Python 3.15.0 (64-bit AMD64), Node.js v24.22.1, FastAPI 0.115, NetworkX 3.4
Test Harness & Telemetry	pytest 9.1.1, pytest-cov 6.0, coverage.py 7.6, anyio 4.15, Node test runner
Browsers Verified	Google Chrome 131, Microsoft Edge 131 desktop, responsive mobile viewports

2.1 Exit Criteria Evaluation (Baseline Quality Tier)

Evaluation Criterion	Target Metric	Achieved Result	Status
Functional Requirements Coverage	**100%**	**100% (43 / 43 FRs covered)**	**Pass**
P1 Critical Test Pass Rate	**100%**	**100% (52 / 52 P1 cases passed)**	**Pass**
Core Backend Code Coverage	>= 85%	**98% (523 / 536 statements)**	**Pass**
Open Severity 1 (S1) Defects	0	0 (DEF-01 resolved in 141fe65)	**Pass**
Dual-Engine Mathematical Parity	**100%**	**100% Identical Output (94.7%)**	**Pass**

3. Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix maps core functional requirements to test cases and empirical execution outcomes.

Req ID	Functional Requirement Description	Test Case IDs	Status
FR-01	User registration with Dhaka coordinate profiles	TC-01, TC-02	**Pass**
FR-03	User deletion with cascading store cleanup	TC-06, TC-07	**Pass**
FR-04	Input validation on user profile coordinates	TC-08, TC-09	**Pass**
FR-06	Skill offer listing registration & category	TC-11, TC-12	**Pass**
FR-08	Prevention of identical offer/want on user	TC-15	**Pass**
FR-10	Case-insensitive trimmed skill matching	TC-18, TC-19	**Pass**
FR-11	MultiDiGraph node synthesis from users	TC-20, TC-21	**Pass**
FR-12	Directed edge insertion matching offer->want	TC-22, TC-23	**Pass**
FR-14	Multi-edge support between users	TC-25	**Pass**
FR-15	Bounded DFS 3-party loop detection (k=3)	TC-26, TC-27	**Pass**
FR-16	Bounded DFS 4-party loop detection (k=4)	TC-28, TC-29	**Pass**
FR-17	Strict cycle length bounding (min_k, max_k)	TC-30, TC-31	**Pass**
FR-18	Rotational canonical cycle deduplication	TC-32, TC-33	**Pass**
FR-19	Zero false-positive cycles on DAG topologies	TC-34, TC-35	**Pass**
FR-22	Haversine geodesic distance calculation	TC-38, TC-39	**Pass**
FR-24	Weekly availability grid modeling (7x24)	TC-42	**Pass**
FR-25	Mutual schedule overlap set intersection	TC-43, TC-44	**Pass**
FR-27	Parametric composite scoring with alpha	TC-46, TC-47	**Pass**
FR-29	Trade proposal initialization & UUID	TC-49, TC-50	**Pass**
FR-31	Unanimous acceptance transition to Locked	TC-53, TC-54	**Pass**
FR-32	Single-veto immediate transition to Rejected	TC-55, TC-56	**Pass**
FR-34	Synthetic Dhaka population generator	TC-59, TC-60	**Pass**
FR-41	Standalone client JavaScript DFS engine	TC-69, TC-70	**Pass**
FR-42	JavaScript Haversine & schedule score parity	TC-71, TC-72	**Pass**
NFR-01	Algorithmic matching < 50ms for N <= 100	PT-01	**Pass**
NFR-04	Dual-engine mathematical score parity (94.7%)	TC-74	**Pass**

4. Execution Results & Code Coverage Telemetry

4.1 Automated Test Execution Summary

Testing Level	Total Runs	Passed	Failed	Pass Rate (%)
Algorithmic Core Unit Tests	35	35	0	100.0%
REST API Gateway Integration Tests	29	29	0	100.0%
Client-Side Standalone Engine Tests	14	14	0	100.0%
Total Automated Test Suite	78	78	0	100.0%

4.2 Backend Statement Code Coverage (98% Achieved)

Module File	Statements	Missing	Coverage (%)	Verified Functional Scope
backend/app.py	140	0	**100%**	FastAPI REST routes & static mounts
backend/models.py	52	0	**100%**	Pydantic domain entities (User, SkillListing)
backend/synthetic_data.py	54	0	**100%**	Dhaka demographic generator & scaling
backend/test_api.py	74	0	**100%**	Full integration suite for REST routes
backend/test_engine.py	70	0	**100%**	Algorithmic DFS bounds & consensus
backend/graph_engine.py	109	7	94%	MultiDiGraph, DFS cycle discovery & scoring
backend/workflow.py	37	6	84%	Atomic proposal state transitions & voting
TOTAL SYSTEM CORE	536	13	**98%**	Full System Verification (Target >= 85%)

4.3 Scalability & Performance Benchmarks

Scale (N)	Edges (M)	Graph Build	Cycle DFS	Total Runtime	Loops Found (k in [3,4])
10 users	24 edges	0.21 ms	0.45 ms	0.66 ms	2 cycles
25 users	86 edges	0.62 ms	1.84 ms	2.46 ms	7 cycles
50 users	210 edges	1.45 ms	8.20 ms	9.65 ms	19 cycles
100 users	534 edges	3.80 ms	38.10 ms	41.90 ms	47 cycles
200 users	1,420 edges	9.10 ms	287.30 ms	296.40 ms	162 cycles
247 users	1,840 edges	12.40 ms	538.60 ms	551.00 ms	241 cycles (capped at 100)

5. Defect Log & Main Bug Fixes

ID	Req Ref	Problem Description	Sev	Status	Resolution Reference
DEF-01	FR-42	Client cycle scores showed static 80.0% due to dummy constants.	S1	**Closed**	Commit 141fe65 (Haversine & schedule port)
DEF-02	FR-18	Duplicate rotational cycles returned when start node was not lowest ID.	S2	**Closed**	Canonical rotational slicing in DFS
DEF-03	FR-32	Proposal remained Accepting after a member rejected.	S2	**Closed**	Single-veto short-circuit to Rejected
DEF-04	FR-04	User creation accepted coordinates outside Bangladesh.	S2	**Closed**	Pydantic coordinate validator
DEF-05	FR-43	Alpha slider updated label but did not re-rank cycles dynamically.	S3	**Closed**	Real-time input event re-sorting
DEF-06	FR-21	Dense synthetic graphs caused memory spikes in DFS.	S3	**Closed**	Depth pruning & 100 cycle limit
DEF-07	FR-10	Skill matching failed on mixed case (e.g. python vs Python).	S3	**Closed**	Lowercase string normalization
DEF-08	FR-44	Vis.js network nodes collapsed on small viewports.	S4	**Closed**	ForceAtlas2 springLength tuning

5.1 Root Cause Analysis of Primary Bug (DEF-01)

In frontend/js/app.js, findClientCycles() contained static placeholder constants (proxScore = 0.85, overlapScore = 0.75). With alpha = 0.5, (0.5 * 0.85) + (0.5 * 0.75) = 0.80 (80.0%) across every cycle. We resolved this by implementing haversineDistanceKm and calculateScheduleOverlapHours in JavaScript, matching the Python backend exactly (Cycle 1: 94.7%, Cycle 2: 69.8%).

6. Individual Contributions & Sign-off

Member Name	Student ID	Assigned Testing Role & Responsibilities	Found	Fixed
Ahammed Abdali Khan	0432320005101118	Bounded DFS verification, Haversine math, client scoring parity, test architecture	4	4
Sumaia Bintey Ismail	0432320005101103	Consensus testing, schedule overlap grid, integration test harness, documentation	3	3
Alin	0432320005101131	Synthetic benchmarks, REST API contracts, negative security tests, QA review	1	1

Release Recommendation: RECOMMENDATION: PROCEED WITH RELEASE (v1.0.1-test). The system demonstrates exceptional algorithmic stability, 100% automated test pass rate across 78 test cases, 98% core backend statement coverage, and zero open critical defects.

Declaration: We declare that this report accurately describes the comprehensive software testing conducted on RoundTable Exchange (RTE) version v1.0.1-test (commit 141fe65). All test cases, coverage telemetry, and performance benchmarks are authentic and reproducible directly from the repository.